Directional bias of a single polarized cell under confinement

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Abstract

Chiral patterns have been observed in various processes from swirling bacterial colonies to tissue morphogenesis and cytoskeletal organization, yet the physical mechanisms underlying chiral cell motion remain poorly understood. Motivated by experiments demonstrating directional bias in the circular motion of confined cells, we use the tools of dynamical systems analysis with computer simulations to identify minimal intrinsic and extrinsic mechanisms capable of generating persistent biased migration. The dynamical systems framework reveals a common organizing principle: directional bias emerges through changes in the stability and/or basins of attraction of the clockwise and counter-clockwise motility states. We find four distinct routes to such bias. First, intrinsic torque in a polarized cytoskeleton can be spatially integrated to produce biased circular motion. Second, anisotropic cell–substrate friction can generate directional preference when reduced friction along the polarity axis is coupled to a directional offset. Third, a chiral wall-alignment response can also produce a persistent directional preference. Finally, substrate patterns that break mirror symmetry, such as dextral or sinistral ridges and troughs, can likewise bias rotational direction. Together, these mechanisms yield distinct, testable predictions and suggest a unifying lens for experimental interrogation of cellular chirality and the design of synthetic systems with programmable chiral motion.

Keywords: motility; directional bias; biophysical modeling; dynamical system analysis; cell biology.

I. INTRODUCTION

From embryonic development to cell migration, the establishment of spatial asymmetries is a fundamental biological process. One manifestation of such asymmetry is chirality — the geometric property of an object that lacks mirror symmetry. Cellular chirality has been observed in diverse settings, including cell migration [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12], alignment on micropatterns [13, 14, 15, 16, 17, 18], cytoskeletal organization [19, 20, 13], cell morphology [9, 21, 22], and multicellular tissues [23, 24]. Although chirality is often linked to cytoskeletal organization, environmental factors such as confinement geometry [25], substrate patterning [16, 17, 18], and external fields [26, 27, 28, 27, 29] can also bias migration. Yet the physical mechanisms that generate persistent directional bias remain poorly understood. Determining how chirality emerges at the single-cell level is therefore a necessary step toward explaining how left–right asymmetries arise across biological scales.

To investigate the physical origins of directional biases, we focus on a minimal model of the migration of a polarized active cell within a disk-shaped confinement. Exploiting the simplicity of the modeling framework, we combine computational simulations with dynamical systems analysis to identify the minimal intrinsic and extrinsic mechanisms capable of generating persistent chiral motion. Building on our previous work on confined cell doublets [30], we show that multiple symmetry-breaking mechanisms can produce robust directional bias, including intrinsic cytoskeletal chirality, anisotropic friction, substrate patterning, and chiral wall interactions. Despite their apparent differences, these mechanisms share a common design principle: persistent chiral motion emerges from an offset between the body-frame polarity axis and the lab-frame axis governing cell–substrate interactions. These mechanisms suggest concrete strategies for experimentally probing the origins of cellular chirality while informing the design of active and living systems with controllable directional behavior.

The rest of the paper is organized as follows. In Sec. II we lay out the biophysical model and its assumptions. Throughout, we follow a single observable: the split of cells between clockwise (CW) and counter-clockwise (CCW) rotation, and we ask what makes that split even, biased, or fully one-sided. The answer depends on one property of the underlying dynamics: whether the cell circulates without ever settling (a conservative system) or settles into a preferred state (a dissipative system). In Sec. III.A, the system is conservative and has no built-in handedness, giving an even CW/CCW split that no parameter can tip. In Sec. III.B, an intrinsic torque in the protrusion dynamics gives the cell a directional preference but leaves the dynamics conservative, which tilts the CW/CCW split toward one side. Sections III.C and III.D make the dynamics dissipative by introducing a chiral offset into one of the two cues that steer the cell's polarity. These two cues are its protrusion direction and the inward direction due to the confining geometry; an anisotropic friction with the substrate underneath selects an "easy" direction of migration, independent of where the cell starts. In Sec. III.E, an adhesive spot in the middle of the confining geometry also makes the dynamics dissipative but without imposing handedness: the CW/CCW split stays even, but now a bistability emerges that can produce tunable chirality when coupled with another mechanism. Finally, in Sec. III.F, we extend the anisotropic-friction mechanism to patterned substrates, where the preferred migration axis is fixed in the laboratory frame. Because these systems no longer admit the same low-dimensional description, we rely on simulations and show that only patterns lacking mirror symmetry generate directional bias. More broadly, our analysis reveals how diverse biological and physical mechanisms can be understood within a common dynamical framework, where chirality emerges through changes in the stability and accessibility of competing motility states.

II. GENERIC DESCRIPTION OF AN ACTIVE PARTICLE IN CONFINEMENT

We propose a minimal mathematical model to capture the interplay between cell polarity, contractility, and spatial confinement (Fig. 1). In our model, the cell is represented by the two-dimensional position of its center-of-mass (ignoring spinning), and further detailed morphology, such as the cell membrane, nucleus, and actomyosin cytoskeleton, is ignored.

Taking the overdamped (or viscous dominated) approximation to a physical system, a cell at location $\mathbf{x}(t) = R(\cos\theta, \sin\theta)^T$ moves with velocity $\mathbf{v}(t)$ according to its local force balance:

$$\sum \mathbf{f}(\mathbf{x}(t), t) = 0 \Rightarrow \mathbf{v} = \dot{\mathbf{x}} = \mathbb{M}\mathbf{f}_{\text{polarity}}, \tag{1}$$

where $\mathbb{M}$ is the mobility matrix. In general, this matrix can capture anisotropies, hydrodynamic interactions, or spatial heterogeneities in the substrate. Here, however, we assume $\mathbb{M} = \frac{1}{\xi}\mathbb{I}$, implying that velocities are directly proportional to applied forces with ξ as the viscous drag coefficient. This is equivalent to elastic cell-matrix interactions under conditions of high dissociation rates of these adhesion bonds [31]. The forces acting on the point are: (1) frictional drag force between the cell and the surface underneath ($\mathbf{f}_{\text{drag}} = \xi\dot{\mathbf{x}}$), and (2) active polarity force arising from front-rear signaling of actin-based protrusions and myosin-based contraction ($\mathbf{f}_{\text{polarity}}$).

Cell polarity. Migrating cells have an underlying chemical polarization, indicating the areas of the cell that are likely to protrude ("front-like") and those likely to contract ("rear-like") [32]. This can include asymmetric distribution of Rho GTPases, with Rac1 activity driving the cell front through lamellipodial extensions, and RhoA promoting myosin contractility in the rear. Rather than explicitly modeling the dynamics of one or more Rho GTPases [33, 34, 35], we summarize cell polarity with a single spatiotemporal motility force $\mathbf{f}_{\text{polarity}}$. This force is a vector with direction ϕ, initially chosen randomly, and magnitude γ_{pol}, a model parameter:

$$\mathbf{f}_{\text{polarity}} = \gamma_{\text{pol}}\hat{\mathbf{p}} = \gamma_{\text{pol}}\left(\cos\phi, \sin\phi\right)^T, \tag{2}$$

and over time evolves via the mechanism of velocity alignment [36]:

$$\dot{\phi} = \frac{|\mathbf{v}|}{V_0}\frac{\sin\left(\theta_V - \phi\right)}{\tau_{\text{VA}}}. \tag{3}$$

Here, $\hat{\mathbf{p}}$ is the unit vector with direction ϕ, θ_V is the angle of the polar parameterization of the velocity vector $\mathbf{v}$ and computed every time step as $\arctan\left(v_y/v_x\right)$, τ_{VA} sets the orientational persistence timescale, and $V_0 = \gamma_{\text{pol}}/\xi$. A long timescale, $\tau_{\text{VA}} \gg 1$, would imply the cell's orientation is insensitive to its own velocity direction and moves with high persistence in the initial polarity direction. This form is in the same spirit as the one suggested originally by [37] and later adapted by [36] and others.

Spatial confinement. To ensure that the cell remains geometrically constrained to the fibronectin-coated micropattern, we incorporate confinement through a reorientation of the polarity angle in Eq. (3):

$$\dot{\phi} = \frac{|\mathbf{v}|}{V_0}\frac{\sin\left(\theta_V - \phi\right)}{\tau_{\text{VA}}} + \frac{\sin\left(\theta_W - \phi\right)}{\tau_W} \tag{4}$$

where $\theta_W = \pi + \theta$ is the angle pointing from the cell's position toward the center of the domain.

This implicit formulation captures how cells with finite spatial extent continuously sense the domain boundaries through their cytoskeleton. Since the polarization represents an averaged cytoskeletal response, the reorientation mechanism naturally accounts for the cell's distributed sensing of geometric constraints without requiring an artificial radial potential that switches on and off at a specific threshold. In the absence of such a confinement effect, the cells engage in persistent directional motion (Fig. S1A).

Quantification of rotational motion by the winding number. To quantify the rotational behavior of trajectories, we use the position angle $\theta(t)$ and its instantaneous angular velocity $\omega(t) \equiv \dot{\theta}$. For a trajectory with $R(t) > 0$ on $[0, T]$, the *winding number* is

$$n(T) = \frac{1}{2\pi}\int_0^T \omega(t)\,dt = \frac{1}{2\pi}\left(\theta(T) - \theta(0)\right). \tag{5}$$

To accurately compute cumulative rotation, angular increments are unwrapped to account for the periodicity of $\theta \in [-\pi, \pi]$, ensuring that full revolutions are retained in the winding number calculation. A trajectory has net CCW rotation if $n(T) > 0$, net CW rotation if $n(T) < 0$, and no net rotation if $n(T) = 0$.

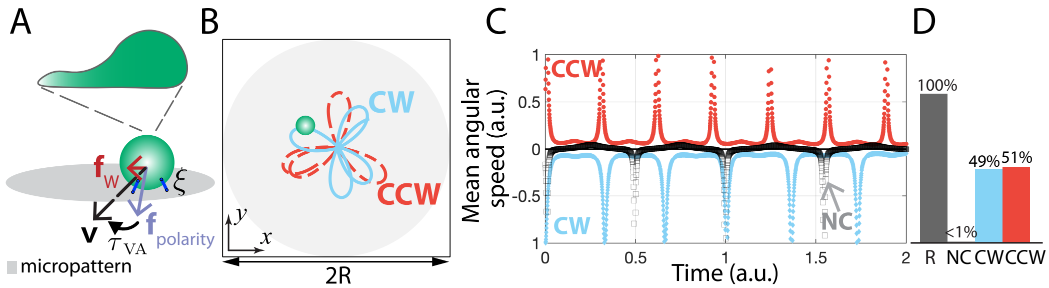

Figure 1: Single migrating cell model and its dynamics. (A) Singlet model schematic. (B) Sample CW and CCW trajectories over one rotational cycle. (C) Angular velocity over 2 arbitrary time units for cells that move non-coherently by switching directionality (black), rotate CW (blue) and CCW (red). (D) Averaged number of rotating singlets and their distribution into the three states: CCW (51%), CW (49%), and NC ($< 0.5\%$) out of 3200 model simulations.

III. RESULTS

Movement in confinement produces a conserved system with reflection-symmetry

To probe the system response, we place a single cell in the center of the disk-shaped micropattern, induce polarization in a random direction, and allow the system to evolve according to Eqs. (1), (2), and (4) for over 3 full rotation cycles (Fig. 1A). Fig. 1B plots one rotational cycle, in either direction, as the singlet navigates the confining geometry in our simulation. Notably, we find that the motion of our default cell displays a petal-like pattern unlike the circular paths observed with cell doublets by the same model [30]. Along with trajectories, we plot the angular speed for several initializations and the averaged behavior over 3,200 simulations.[1] (Fig. 1C-D). Equations of motion are solved numerically using the Forward Euler integration scheme, and parameter values are provided in Table S1.

Cells display a variety of motility patterns (Fig. S2), including cells moving persistently in one direction, either clockwise (CW) or counterclockwise (CCW), or switching direction (Fig. 1C-D, Movie 1), thus not moving coherently (NC). The marginal case of non-coherent movement appears at a very low frequency. To classify the emergent behavior, winding number is used – non-coherent movement occurs if the cell switches directionality over some arbitrary percentage.[2], with clockwise for negative values or counterclockwise for positive values of the winding number. This means that our model, at its default parameters, predicts three-state behavior, with nearly all cells persistently rotating either clockwise or counterclockwise with equal transition rates (Fig. 1E, Fig. 2A).

To verify the computational simulation results, we transform to polar coordinates where $\mathbf{x} = R(\cos\theta, \sin\theta)^T$ and introduce the phase-lag $\Delta\varphi = \phi - \theta$, quantifying the angular offset between polarity and positional angle. From the equation of motion, we find that $\mathbf{v} = \gamma_{\text{pol}}\hat{\mathbf{p}}/\xi$, and consequently the velocity is parallel to the polarity, so that $\theta_V = \phi$. The polarity equation reduces to $\dot{\phi} = \sin(\Delta\phi)/\tau_W$ since $\sin(\theta_W - \phi) = \sin(\theta + \pi - \phi) = \sin(\pi - \Delta\phi) = \sin\Delta\phi$. By re-writing the equation of motion in polar coordinates (Appendix A), and subtracting $\dot{\theta}$ from $\dot{\phi}$, one can derive the closed system on $(R, \Delta\varphi)$:

$$\dot{R} = \frac{\gamma_{\text{pol}}}{\xi}\cos\Delta\varphi, \quad \dot{\Delta\varphi} = \sin\Delta\varphi\left(\frac{1}{\tau_W} - \frac{\gamma_{\text{pol}}}{\xi R}\right). \tag{6}$$

Phase plane analysis of this reduced system explains the equal CW/CCW split (Fig. 2B, Appendix B). The phase space divides into two regions, an upper half in which the cell migrates counter-clockwise and a lower half in which it migrates in the clockwise direction, set apart by invariant dividing lines that the dynamics never crosses. Therefore, a cell retains whichever direction it starts with, and each region is organized around a single center, the only state in which R does not oscillate (Fig. 2C).

As suggested by the closed trajectories of the reduced 2D plane, we find that the system admits a conserved Hamiltonian quantity, $H(R, \Delta\varphi)$ (Appendix B). Trajectories lie on the level sets of H (Fig. 2B). Moreover, the reduced system is invariant under reflection; every trajectory $(R, \Delta\varphi)$ has a mirror image trajectory $(R, -\Delta\varphi)$ (Appendix B). Since the model parameters are scalar quantities that are unchanged by this reflection, no parameter variation can break the equal CW/CCW split. We confirmed this result numerically with parameter sweeps (Fig. S1B).

Intrinsic torque breaks direction symmetry

In pursuit of symmetry breaking in the system, and emergence of directionality preference, we probed the model's response to the presence of intrinsic bias in the polarity machinery. We extended the polarity model in Eq. (4) by

[1] Without making assumptions about the underlying ratio of a binary proportion ($p = 0.5$), and assuming a margin of error $E = \pm 3\%$, we require $n = (Z^2 \times p \times (1-p))/E^2 = 1,068$ samples for 95% confidence ($Z = 1.96$). Assuming that the lowest ratio of rotation to non-coherent movement never goes below 30%, we conclude that $3,200$ samples yield 95% confidence with error intervals of $\pm 3\%$. Any cases where the ratio of rotation to non-coherent movement falls below 40% will be classified as statistically unreliable.

[2] Over 20% unidirectional is considered coherent rotation.

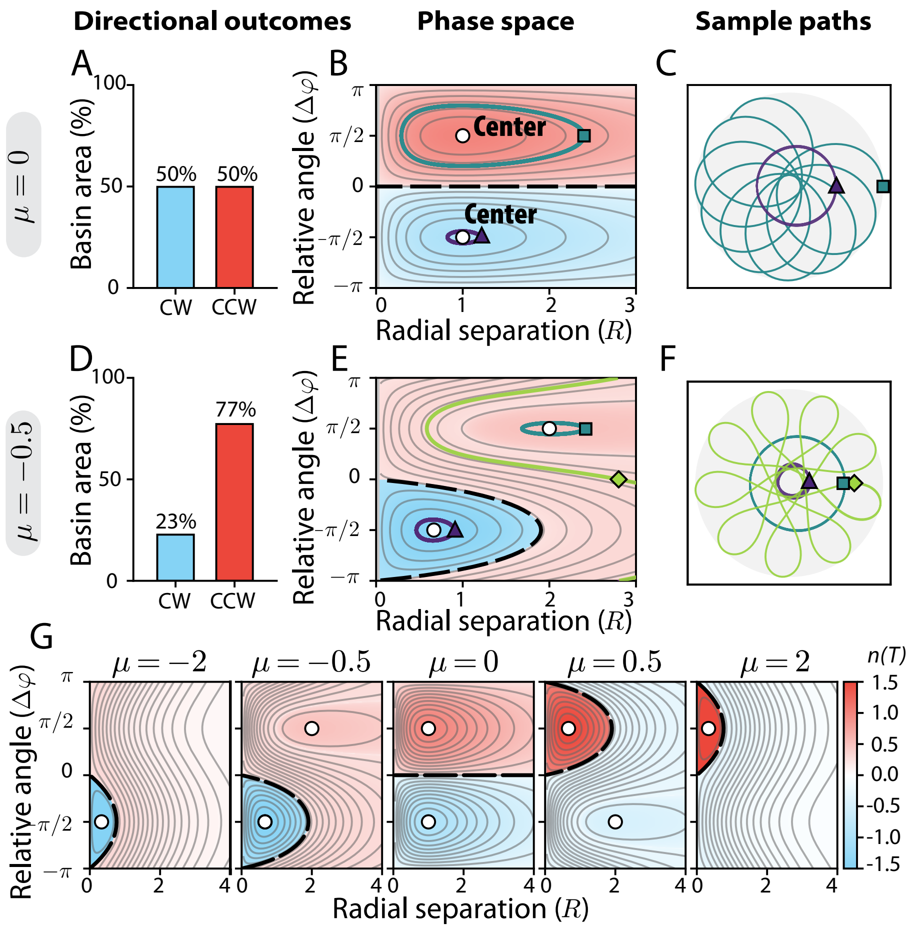

Figure 2: Directionality bias can either persist or disappear in confining geometries when cells have an added intrinsic torque in their polarity orientation. (A, D) CW/CCW motility outcomes for $\mu = 0$ and $\mu = -0.5$. (B, E) Phase portraits in $(R, \Delta\varphi)$ colored by the time-averaged winding number $n(T)$, with representative trajectories (gray and colored lines), flow direction (arrows), invariant lines (dashed black), and neutral centers (white circles). (C, F) Example cell trajectories corresponding to selected initial conditions in (B, E). (G) Phase space scanning the intrinsic bias parameter μ from -2 to 2, colored by $n(T)$, shows the breakdown of CW/CCW rotational states.

adding an intrinsic torque μ:

$$\dot{\phi} = \frac{1}{\tau_{\mathrm{VA}}} \frac{|\mathbf{v}|}{V_0} \left[\sin\left(\theta_V - \phi\right) + \mu \right] + \frac{1}{\tau_{\mathrm{W}}} \sin\left(\theta_W - \phi\right). \tag{7}$$

Here, $\mu < 0$ leads to CW rotations, and $\mu > 0$ to CCW rotations. For example, for $\mu = -0.5$, this intrinsic bias in the evolution of the polarity angle can translate into a bias towards CW rotational movement (Fig. 2D, trajectories in Fig. S3).

Phase plane analysis confirms the model simulations: for $\mu \neq 0$ the direction symmetry is broken (Fig. 2D-E, Appendix C). Importantly, the bias does this without destroying the conserved structure of the unbiased model: the reduced system still admits a conserved quantity so the two rotational states remain neutrally stable centers. What the intrinsic bias does change is the balance between the two regions: each center persists but shifts, so that for $\mu < 0$ the clockwise center tightens into a smaller orbit (Fig. 2F) and claims a larger share of initial conditions Fig. 2G), while counter-clockwise center widens and claims fewer initial conditions. Every trajectory retains a definite, signed winding number, clockwise or counter-clockwise.

As the intrinsic bias grows further, it pushes the disfavored center into ever-wider orbits, until a critical strength ($|\mu| = 1/\tau_W$), where its radius diverges and it collides with a saddle point sitting at infinity, annihilating in a saddle-center bifurcation (see Appendix C). Beyond this point a single rotational state survives, CCW for $\mu > 1/\tau_W$ and CW for $\mu < -1/\tau_W$, and the cell turns one way regardless of how it starts. An intrinsic bias ($\mu \neq 0$) is therefore both necessary and sufficient to produce directed rotation in a single cell. In previous work [30], we found that this type of dynamics, extended to cell pairs, can produce richer behavior, including regimes where doublets overcome an intrinsic clockwise bias to rotate predominantly counter-clockwise.

Lower frictional drag along an offset direction creates directional bias

Above, we explored the possibility that a directional preference in cellular motility can arise from an intrinsic torque in the cellular response (polarity), but how that can emerge remains opaque. To illustrate one possible underlying mechanism, we allow cell-substrate frictional drag to be anisotropic: moving in one direction is easier than moving across it. The anisotropy could be introduced through several distinct possibilities, including intrinsic cytoskeletal alignment, confinement-induced shape anisotropy, or patterning of ridges and troughs directly onto the surface (see Section III.F). We show that a frictional anisotropy in the model can produce a preferred migratory direction through a mechanism that is dissipative rather than the conserved dynamical system in the previous section (Fig. 3).

First, the intrinsic bias is removed from Eq. (7). To model anisotropic friction, we introduce an "easy" axis, $\hat{\mathbf{e}}_{\parallel}$; we assume that motion parallel to this easy axis is governed by the friction coefficient $\xi_{\parallel}$, and $\xi_{\perp}$ in the perpendicular direction. The ratio of these coefficients controls migratory persistence anisotropy ($\xi_{\perp} > \xi_{\parallel}$) by promoting alignment with the easy direction. The mobility matrix tensor is

$$\mathbb{M} = \frac{1}{\xi_{\perp}}\left(I + \alpha \hat{\mathbf{e}}_{\parallel}\hat{\mathbf{e}}_{\parallel}^{T}\right), \quad \alpha = \frac{\xi_{\perp} - \xi_{\parallel}}{\xi_{\parallel}} > -1. \tag{8}$$

The α term introduces an orientation-dependent coupling between radial and angular dynamics. In the limit that $\alpha = 0$ ($\xi = \xi_{\perp} = \xi_{\parallel}$), Eq. (8) reduces to $\mathbb{M} = \frac{1}{\xi}\mathbb{I}$ as expected. Note that for $\alpha \leq -1$ the mobility matrix is no longer positive-definite and we exclude this case.

If we assume that easy axis is lagging the polarity by a fixed offset angle δ, $\hat{\mathbf{e}}_{\parallel}(\phi) = (\cos(\phi+\delta), \sin(\phi+\delta))^{T}$, one can show that the update velocity becomes

$$\mathbf{v} = \frac{\gamma_{\text{pol}}}{\xi_{\perp}}\left[(1+\alpha\cos^{2}\delta)\hat{\mathbf{p}} + \frac{\alpha}{2}\sin(2\delta)\hat{\mathbf{p}}_{\perp}\right]. \tag{9}$$

Here, $\hat{\mathbf{p}} = \mathbf{f}_{\text{polarity}}/\gamma_{\text{pol}} = (\cos\phi, \sin\phi)^{T}$ as in Eq. (2), and $\hat{\mathbf{p}}_{\perp} = (-\sin\phi, \cos\phi)^{T}$ (Appendix D). We find that the cell's velocity is not perfectly aligned with its polarity vector due to the presence of the second term in Eq. (9). Instead, the cell picks up a small sideways component (proportional to $\sin(2\delta)$), meaning it "slips" slightly sideways as it moves. Consequently, the velocity alignment term does not vanish and remains as a constant applied torque to the system: $\dot{\Delta\varphi} = \frac{1}{\tau_W}\sin(\theta_W - \phi) + \mu_{\text{eff}}$, where $\mu_{\text{eff}} = \alpha\sin(2\delta)/2\tau_{\text{VA}}$.

For $\delta = 0$, the system simplifies to the unbiased case, with no intrinsic directional preference, as in Eq. (6). However, a nonzero offset ($\delta \neq 0$) does more than add a bias; it changes the dynamics qualitatively. The offset destroys the Hamiltonian dynamics and makes the reduced $(R, \Delta\varphi)$ flow non-conservative. For moderate anisotropy the two neutrally stable centers become hyperbolic spirals, one stable and one unstable; for stronger anisotropy the sink instead escapes to infinity, a nonphysical runaway we set aside (Fig. 3, Appendix D). These equilibria are spirals, never saddles ($\det J > 0$), and their stability is set entirely by the sign of $\alpha\sin(2\delta)$. We find that the resulting bias produces all-or-nothing CW/CCW outcomes rather than a tilting of outcome probabilities; for example, if $\alpha > 0$ (favored alignment with the easy axis), a positive δ stabilizes the clockwise state, and destabilizes the counter-clockwise one (top right, Fig. 3).

Because the stable spiral is a global attractor for moderate anisotropy, this directional preference is robust to initial conditions: every initialization of the singlet converges to the same rotational state, which our computational simulations confirm (all tested initial conditions produce persistent clockwise rotation; inset Fig. 3). Beyond the threshold anisotropy the interior attractor is lost and the reduced trajectory instead spirals outward to unbounded radius (Appendix D). Such an escape is unphysical for a cell confined to the disc, so we do not consider it further. The dissipation also reshapes the orbits, replacing the closed petal-like trajectories of the conservative model with spirals that settle onto a singular circular orbit (Fig. S4).

Taken together, these results demonstrate that reduced cell-substrate frictional resistance along the polarity axis, combined with an offset arising from chiral cytoskeletal organization, is sufficient to generate a directional bias in cell movement. The next mechanism breaks symmetry in the same dissipative way, but through the cell's interaction with the confining geometry.

Chiral alignment due to confinement selects a single motility direction

Another mechanism that breaks the reflection symmetry of Eqs. (1)-(4) is through confinement effects. Here, we return to isotropic cell-substrate friction, so the cell's velocity is again parallel to the polarity ($\theta_V = \phi$) and the velocity alignment term drops out; the symmetry breaking comes entirely from confinement. We offset the alignment of polarity direction by a fixed angle χ, replacing $\sin(\theta_W - \phi)$ with $\sin(\theta_W - \phi + \chi)$. Together with $\theta_W = \theta + \pi$, the confinement term becomes $\sin(\Delta\varphi - \chi)/\tau_W$, and the closed system on $(R, \Delta\varphi)$ is

$$\dot{R} = \frac{\gamma_{\text{pol}}}{\xi}\cos\Delta\varphi, \quad \dot{\Delta\varphi} = \frac{\sin(\Delta\varphi - \chi)}{\tau_W} - \frac{\gamma_{\text{pol}}}{\xi R}\sin\Delta\varphi. \tag{10}$$

Notably, the symmetric flow in Eq. (6) has shifted by χ.

Expanding the chiral term in Eq. (10) gives $\frac{1}{\tau_W}\sin(\Delta\varphi - \chi) = \frac{1}{\tau_W}\cos\chi\sin\Delta\varphi - \frac{1}{\tau_W}\sin\chi\cos\Delta\varphi$. The second term in the expression breaks reflection symmetry ($\Delta\varphi \to -\Delta\varphi$), but unlike the constant torque μ, it carries a $\cos\Delta\varphi$ factor, so it vanishes at the two equilibria $\Delta\varphi = \pm\pi/2$. This makes it tempting to conclude that χ produces no bias, and indeed χ does not shift the steady states. Instead, exactly as for the offset from the polarity axis, it renders the system dissipative and splits the stability of the two states: centers become hyperbolic spirals,

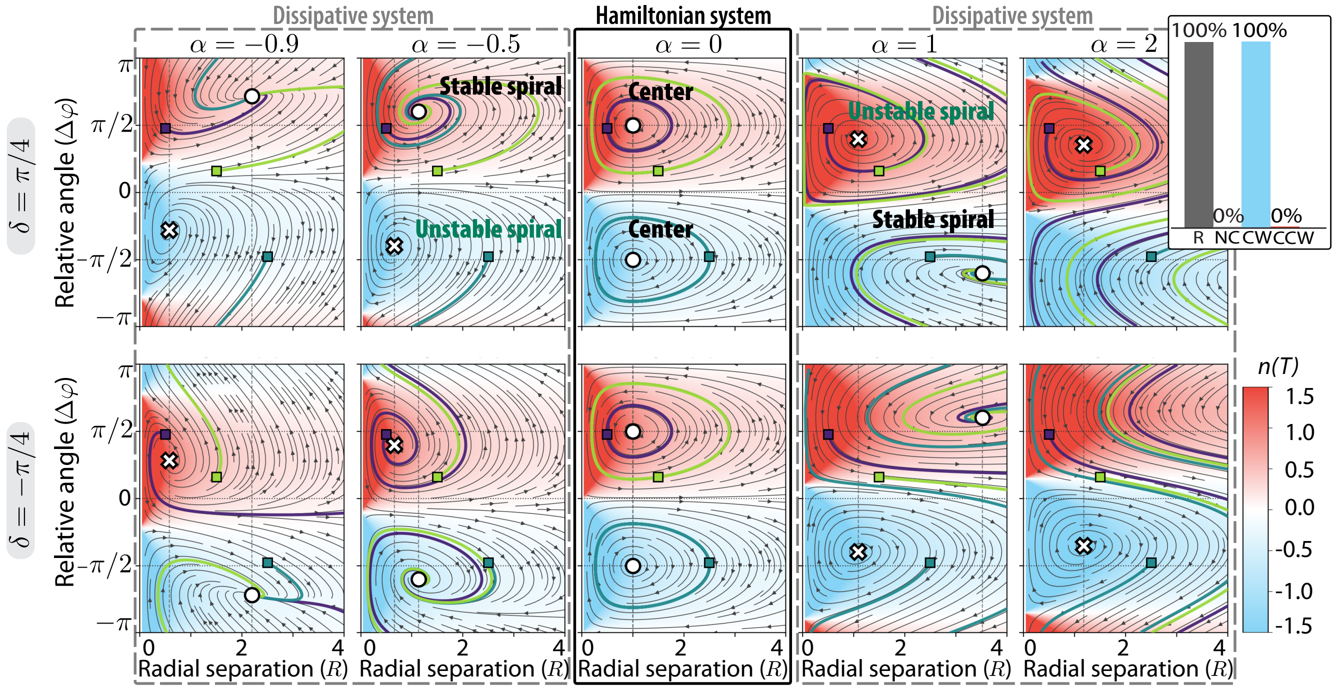

Figure 3: Phase portraits illustrate the transition from conserved (Hamiltonian) to dissipative dynamics and the resulting changes in the stability of CW and CCW equilibria with offset angle δ and frictional anisotropy α. Top: $\delta = \pi/4$; bottom: $\delta = -\pi/4$. From left to right, α increases from negative to positive values. Phase space $(R, \Delta\varphi)$ is colored by the time-averaged winding number $n(T)$. White circles denote stable (or neutrally stable) equilibria, white crosses unstable equilibria, gray curves representative trajectories, and arrows the flow direction. Inset: fraction of rotating singlets and their distribution among motility states for $\delta = \pi/4$ and $\alpha > 0$.

one stable and one unstable, at a common rescaled radius (Appendix E, Fig. S5). The stability of the equilibria, and hence the handedness of the motility, is dictated by the sign of parameter χ. Notably, chiral confinement produces the same all-or-nothing bias as the polarity axis offset, by splitting the stability of the two rotating states rather than displacing them.

A center adhesive spot turns intrinsic chirality mechanisms into tunable bias

In an effort Ferencto introduce different cell-matrix adhesion topology, we first consider a radial cell-matrix force at the center of the disk-shaped pattern (Fig. 4A, pattern (ii)) in Eq. (1): $\mathbf{f} = \mathbf{f}_{\text{polarity}} + \mathbf{f}_{\text{anchor}} = \gamma_{\text{pol}}\hat{\mathbf{p}} + f(R)\hat{\mathbf{r}}$, where $f(R) = k_0 R$. The force is radially oriented and is equivalent to an elastic spring tethering the cell to the origin; by controlling the sign of its coefficient k_0, the force can be either repelling ($k_0 < 0$) or attracting ($k_0 > 0$). The reduced dynamics are given by

$$\begin{aligned} \dot{R} &= \frac{\gamma_{\text{pol}}}{\xi}\cos\Delta\varphi + \frac{f(R)}{\xi}, \\ \dot{\Delta\varphi} &= \sin\Delta\varphi\left[\frac{1}{\tau_W} - \frac{\gamma_{\text{pol}}}{\xi R} - \frac{f(R)}{\gamma_{\text{pol}}\tau_{\text{VA}}}\right]. \end{aligned} \tag{11}$$

In the absence of the additional radial force, Eq. (6) is recovered. For $f \neq 0$, the cell's velocity is no longer parallel to its polarity direction, so, as with the polarity offset, the flow becomes dissipative and the neutrally stable centers become spirals. Yet, similar to the default cell on a disk-shaped micropattern, the outcomes remain unbiased, with an equal likelihood of CW and CCW motion, as evidenced by the equally sized, well-defined basins of attraction of the two steady states (Fig. 4D(i)). We conclude that the presence of an adhesive dot alone does not produce bias in the directional motion of the singlet cells exploring the confined geometry.

What the presence of the additional radial force enables is a tunable bias in the chirality mechanisms we have discovered so far. Because the two equilibria are not attractors, any symmetry breaking coupling (an intrinsic torque $\mu \neq 0$, a polarity axis offset, $\delta \neq 0$, or a chiral confinement $\chi \neq 0$) tilts the basin boundary toward the disfavored sink and enlarges the favored basin, so the population handedness shifts continuously with the symmetry breaking strength. For example, for the case of a CW intrinsically biased cell ($\mu < 0$), changing the directionality of the anchor (attractive or repulsive) reveals either an overall CW or a CCW directional bias in both the phase space analysis and model simulations (Fig. 4B-D). With an attracting anchor, CCW bias emerges, while a repulsive anchor enhances the intrinsic CW bias.

Dynamical system analysis further shows that a strong enough intrinsic bias (μ) or chiral confinement (χ) removes

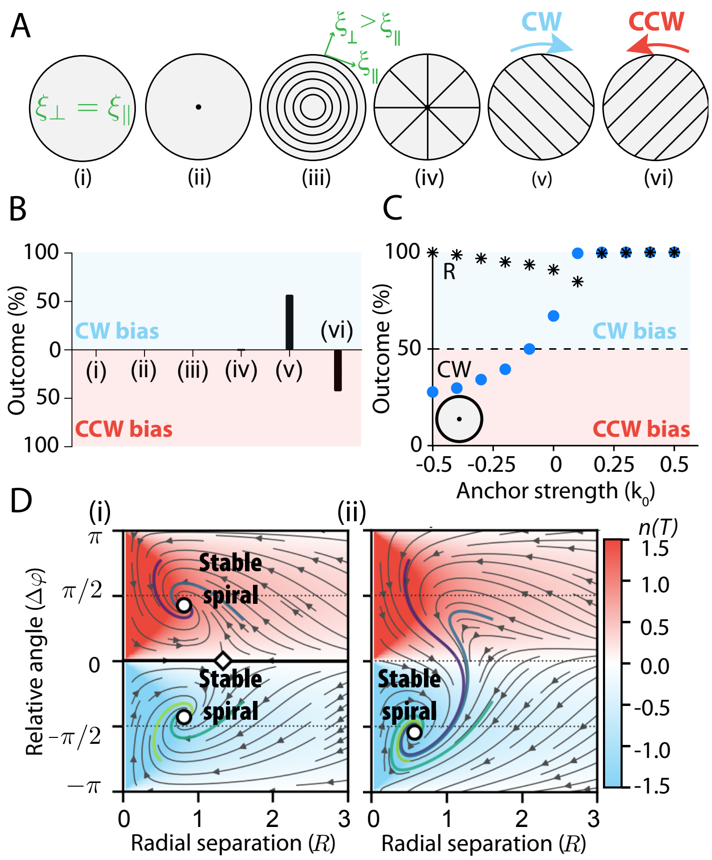

Figure 4: Model predicts directional biases on substrate patterns that lack mirror symmetry. (A) Substrate adhesive patterns. (B) Directionality outcomes showing deviations from equal CW/CCW distribution for substrate patterns in (A). (C) Motility outcomes of the CW-biased cell ($\mu = -0.5$) on pattern (ii) as a function of anchor strength k_0. (D) Phase portraits in $(R, \Delta\varphi)$ for pattern (ii) with a center adhesive dot ($k_0 = 0.5$, $k_W = 1$), for (i) an unbiased cell ($\mu = 0$) and (ii) a CW-biased cell ($\mu = -0.5$), respectively.

the disfavored state and makes the cell one-handed, by one of two routes: the intrinsic bias slides the disfavored sink into the basin boundary until the two annihilate (a saddle-node bifurcation), whereas the chiral confinement drives the disfavored sink unstable (a Hopf bifurcation) (see Appendix F, Fig. S6). The polarity axis offset (anisotropic friction) is the exception: within the physical range its effective torque is bounded and too weak to remove a sink, so the (α, k_0) plane stays bistable and only skews the population. This behavior is illustrated in Fig. 5: under some conditions, cells can rotate in either direction, but sufficiently strong intrinsic bias (μ) or chiral cell-wall interactions (χ) ultimately favor a single direction of rotation.

Substrate patterning as an extrinsic route to directional biases

A final mechanism we report on for attaining a directional bias in confined motility is through patterned substrates. We focus on patterns as depicted in Fig. 4A and compare the CW/CCW split outcomes against the default disk-shaped isotropic pattern (Fig. 4B). In contrast to the previous sections, these patterns do not generally reduce to the same low-dimensional (2D) dynamical system and therefore are not amenable to the same analytical treatment. Nevertheless, their behavior can still be systematically characterized through numerical simulations.

To model a patterned substrate, we assume that there is spatial variation along the easy axis, $\hat{\mathbf{e}}_{\parallel} = g\hat{\mathbf{x}} + h\hat{\mathbf{y}}$, where g, h are components of $\hat{\mathbf{e}}_{\parallel}$ in the lab frame such that $g^2 + h^2 = 1$. A simple computation in 2D gives $\mathbb{M} = \frac{1}{\xi_\perp}\left(I + \alpha\begin{pmatrix} h^2 & -gh \\ -gh & g^2 \end{pmatrix}\right)$, where $\alpha = \frac{\xi_\perp - \xi_\parallel}{\xi_\parallel} > -1$ is defined as before in Eq. (8). In Fig. 4A, substrate patterns (iii) and (iv) are defined by $g = g(\mathbf{r}) = \cos(\theta + \theta_0)\hat{\mathbf{x}}$ and $h = h(\mathbf{r}) = \sin(\theta + \theta_0)\hat{\mathbf{x}}$ with $\theta_0 = \pi/2$ and 0, respectively. Patterns (v) and (vi) utilize $g = 1/\sqrt{2}\,\hat{\mathbf{x}}$ and $h = \mp 1/\sqrt{2}\,\hat{\mathbf{y}}$, respectively. A summary of the averaged singlet motility outcomes is provided in Fig. 4B.

We find that the even distribution in motility direction is disrupted only by patterns that lack mirror symmetry: patterns (v) and (vi) in Fig. 4A. For the case of mirror symmetric patterns (Fig. 4A(iii)-(iv)), further varying α does modulate the number of rotational outcomes, but does not impact the directionality bias (Fig. 4B). For substrate patterns that lack mirror symmetry, cases (v) and (vi), the model predicts the emergence of a directional bias provided that the frictional anisotropy is sufficiently strong (α, Fig. S7A), even in the absence of any additional symmetry-breaking mechanisms. The resulting trajectories retain their characteristic petal-like structure but become increasingly skewed and elongated along the easy frictional direction (Fig. S7B-C). This is because the lower frictional coefficient makes it easier to travel along that direction. As the degree of frictional anisotropy increases, the magnitude of the directional bias grows; however, the probability of observing persistent migration correspondingly decreases (Fig. S7A). Instead, the cells increasingly switch directionality. For the results shown in Fig. 4B, we selected $\alpha = 0.4$ as a representative value that produces a measurable directional bias while maintaining sufficient persistence. Because persistent trajectories become less frequent under these conditions, the number of simulation realizations was increased four-fold to obtain robust statistics. More broadly, the results suggest that spatial patterning of the cellular microenvironment may provide a simple extrinsic mechanism for guiding chiral migration and establishing directional preference in confined cells.

IV. DISCUSSION

Chirality is observed across biological scales, yet it is unknown how microscopic asymmetries converted into robust, macroscopic directional behavior? Here, we identify four minimal routes through which persistent chiral migration can emerge (Fig. 6): (1) an intrinsic torque manifested in the front-rear polarity establishment, (2) a frictional anisotropy along the protrusion direction (with an offset), (3) an intrinsic torque manifested due to geometric confinement, and (4) a frictional anisotropy along physical patterns on the substrate. Although these mechanisms originate from distinct biological or environmental sources, they share a common principle: directional bias arises when local asymmetries are coupled to a reference frame that breaks mirror symmetry.

These identified mechanisms have been reported biologically. Experiments in dHL60 cells exposed to a uniform chemoattractant found a strong left-right bias in spontaneous polarization relative to the centrosome [1]. Because no external directional cue was present, these results suggest an intrinsic chiral asymmetry in the polarity machinery. In our framework, such asymmetry is represented by either an intrinsic torque or an offset angle between the polarity and motility axes, which acts as a symmetry-breaking term and biases the system toward persistent CW or CCW motion. Potential biological origins of such an intrinsic offset include persistent cytoskeletal anisotropy associated with mechanical memory [38] and asymmetries generated by nuclear positioning within the cell [29]. Consistent with our third mechanism for chiral motion, recent work suggests that intrinsically chiral actin organization can bias the orientation of cells relative to tissue boundaries, causing boundary cells to adopt a preferred left- or right-tilted alignment [39]. This framing is congruent with our proposed chiral wall-alignment mechanism: the boundary acts as a reference frame, cells align to it with a handed offset, and that local interaction is sufficient to produce persistent large-scale chirality. Lastly, spatially constrained experiments of cells placed on ridge patterned surfaces [40, 41, 42, 43, 16, 17, 18] support our fourth and final mechanisms for chiral motion.

A recurring theme across our results is that chirality is not solely an intrinsic property of the cell. Instead, environmental interactions can reveal, amplify, suppress, or even reverse an underlying directional preference. Fixed adhesion sites,

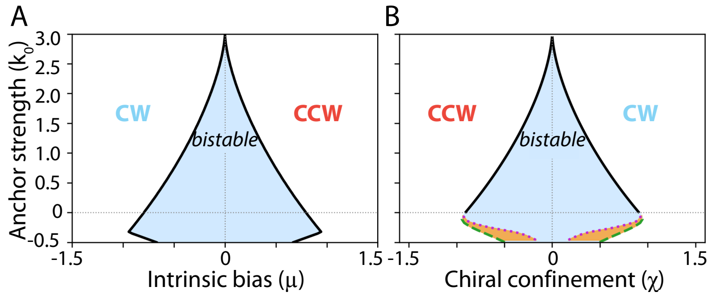

Figure 5: Two-parameter bifurcation diagrams for the single cell on a substrate with a center adhesive spot with (A) intrinsic bias (μ) and (B) chiral cell-wall interactions (χ). In both cases, there is a range of conditions where clockwise (CW) and counterclockwise (CCW) rotation can both occur. As the strength of the intrinsic bias or chiral confinement increases, the system transitions to a regime in which a single rotational direction is preferred.

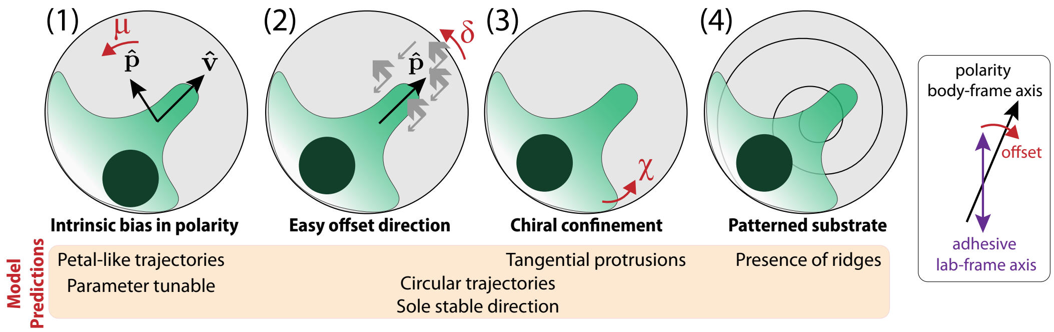

Figure 6: Summary: Identified mechanisms for chiral rotational movement of singlets in confining micropatterns: (1) intrinsic torque μ in polarity direction, (2) offset from the polarity direction through an "easy" frictional direction, (3) chiral confinement forces, and (4) chiral physical patterning on the substrate. Model suggests that chirality in directional individual cell migration emerges with an offset between polarity axis and cell-substrate adhesion axis.

geometric constraints, neighboring cells, or substrate architectures can serve as reference frames that transform latent asymmetries into observable migratory biases. This observation suggests that experimentally measured chirality may reflect an interplay between cellular properties and environmental context rather than either factor alone.

Accompanying computational simulations, the analytical framework reveals a common dynamical organization underlying chiral migration. Across multiple scenarios, the system exhibits competing CW and CCW attractors whose relative stability is tuned by intrinsic and extrinsic symmetry breaking effects. Different mechanisms act by reshaping this dynamical landscape, altering basin sizes, destabilizing equilibria, or transforming conservative dynamics into dissipative ones. Viewed through this lens, diverse manifestations of chirality can be understood as different perturbations of a shared underlying dynamical structure.

Our work provides a unifying framework for understanding how chirality can emerge from the interaction of cellular polarity, mechanics, and environmental structure. By identifying minimal routes to directional bias and the signatures associated with each, the framework offers experimentally testable hypotheses for probing the origins of biological chirality. At the same time, the principles uncovered here may inform the design of synthetic active materials and engineered systems, such as microvessels, that exploit symmetry breaking to achieve controlled chirality.

Competing interests. The authors declare no conflicts of interest.

Data availability. The article is a theoretical study and does not include any empirical data. The Matlab code used to simulate the model is deposited on Zenodo: `https://zenodo.org/records/20753115`.

Use of AI. The authors used AI tools for language editing purposes, and an AI coding assistant (Anthropic's Claude) to help write and refine the Python code used to generate the figures. All final content, arguments, and conclusions remain the sole responsibility of the authors.

Author contributions. Both authors contributed equally in all aspects of this work including conceptualization, formal analysis, methodology, investigation, writing - original draft, writing - review and editing.

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